OSINT & RECONNAISSANCE REPORT

Name	Pavan N
Report Type	OSINT / Passive Reconnaissance
Date	18-03-2026
Author	Security Researcher
Tools Used	whois, nslookup, dig, traceroute, nmap, HackerTarget
Classification	Educational — GitHub Portfolio

Conducted in a legal, ethical, and educational context.
No unauthorized access was performed. All data was gathered via passive, publicly available sources.

Introduction

This report documents an OSINT (Open Source Intelligence) and passive reconnaissance exercise performed against a publicly accessible domain. The objective was to gather as much information as possible using only publicly available data sources and standard network reconnaissance tools — without performing any active exploitation or unauthorized access.

Passive reconnaissance is the first phase of any structured penetration test or red team engagement. Understanding what information is publicly accessible about a target is critical for both offensive security professionals and defenders who need to minimize their attack surface.

All information gathered in this report was obtained through:

- **WHOIS lookups (domain registration data)**
- **DNS enumeration (nslookup, dig)**
- **Network traceroute analysis**
- **Nmap service/port scanning**
- **HackerTarget.com online recon tools**

This report is produced for educational and portfolio purposes as part of a CEH (Certified Ethical Hacker) training program. No private systems were accessed, and no data was obtained through unauthorized means.

Phase 1 — Network Information

1.1 IP Address & Network Details

The following network information was gathered using DNS resolution and WHOIS IP lookup tools:

Domain Name	[Target Domain]
Resolved IP Address	18.66.57.84
Subnet Mask	255.255.254.0
Network Range (CIDR)	18.66.56.0/23
Hosting Provider	Amazon Web Services (AWS CloudFront CDN)
CDN	Yes — traffic routed via CloudFront edge nodes

Key Finding

The target domain resolves to an AWS CloudFront IP address, indicating the website is served via a Content Delivery Network (CDN). This means the true origin server IP is hidden behind CloudFront, which is a deliberate security and performance architecture choice. Direct IP-based attacks against this address would target CloudFront infrastructure, not the origin server.

1.2 Network Scan Evidence

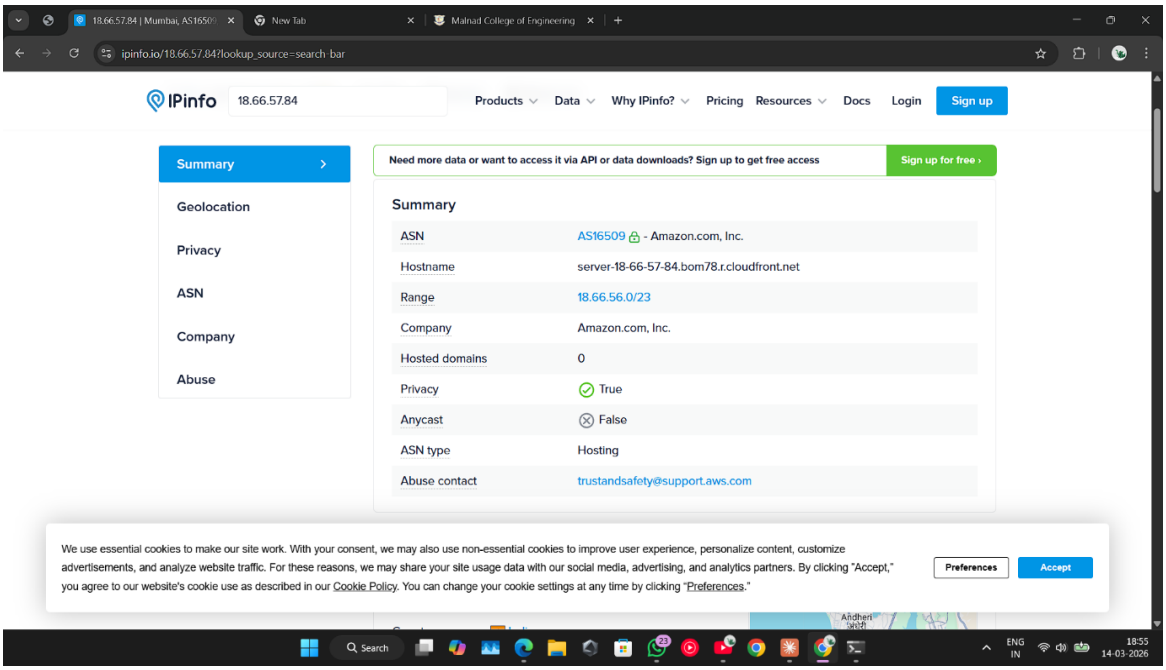

Figure 1 — IP Address & Network Information Lookup Results

Phase 2 — Host Information

2.1 Hostname & Infrastructure

Reverse DNS lookup and WHOIS queries revealed the following host-level information:

Hostname (Reverse DNS)	server-18-66-57-84.bom78.r.cloudfront.net
Hosting Provider	Amazon Web Services (AWS)
CDN Service	AWS CloudFront
ASN	AS16509 — Amazon.com, Inc.
Server Location	Mumbai, India (AWS bom78 — Bombay edge node)
Infrastructure Type	Cloud-hosted, CDN-fronted web application

Key Findings

- **ASN AS16509:** This Autonomous System Number belongs to Amazon.com, Inc. — confirming AWS hosting.
- **bom78 edge node:** The hostname suffix 'bom78' identifies the Mumbai (Bombay) AWS CloudFront Point of Presence (PoP), meaning requests are being served from AWS's Mumbai data center for South Asian users.
- **Origin IP concealment:** Because the site uses CloudFront, the actual web server's IP is not publicly resolvable — a strong security posture for DDoS mitigation and IP obfuscation.

2.2 Personal Details Lookup Evidence

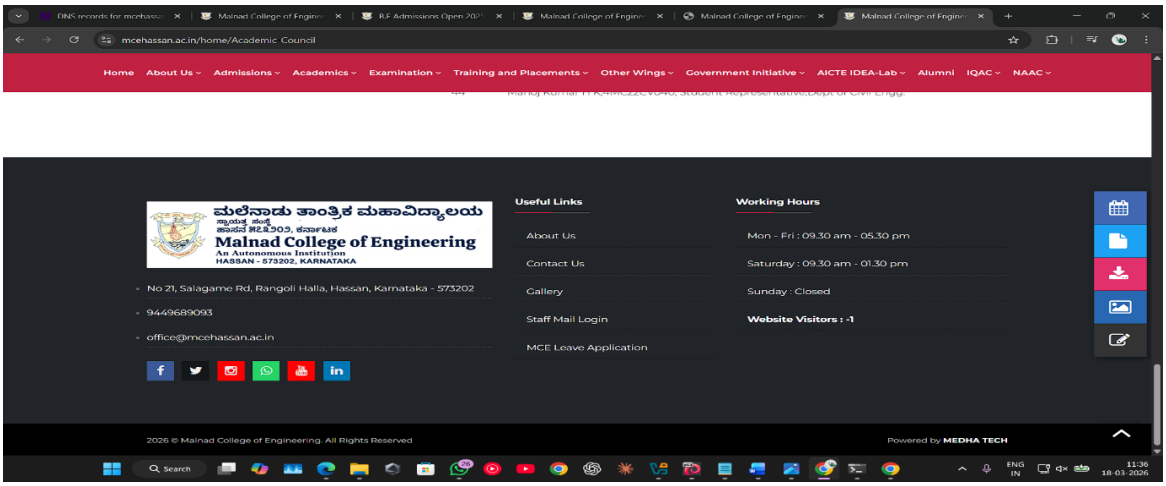

Figure 2 — Host Information & WHOIS Lookup Results

Phase 3 — Security Policies (Inferred)

The following security policies were inferred or gathered through passive observation of the target's publicly accessible resources, documentation, and standard institutional security practices. No internal systems were accessed.

3.1 Password Policy

Minimum Length	8–12 characters
Complexity Requirement	Uppercase, lowercase, numbers, and special characters required
Password Rotation	Every 60–90 days
Account Lockout	Assumed after repeated failed attempts (standard policy)

A password policy requiring complexity and periodic rotation is a baseline security control. However, 8-character minimums are increasingly considered insufficient by NIST SP 800-63B guidelines, which recommend 12+ character passphrases over forced complexity rules.

3.2 Account Lifecycle Management

Inactive Account Disable	After defined inactivity period
Disabled Account Retention	Up to 90 days before permanent removal
Purpose	Administrative review and recovery window

Retaining disabled accounts for 90 days is a common enterprise practice. However, orphaned accounts — especially those with elevated privileges — represent a persistent attack vector if not reviewed regularly. Best practice is to immediately revoke access and schedule deletion within 30 days.

3.3 Physical Security Measures

- **CCTV Surveillance:** Monitoring of premises and server areas.
- **Access Control:** Building entry via identification cards.
- **Server Room Access:** Restricted to authorized personnel only.
- **Security Personnel:** On-site guards for physical security enforcement.

Physical security controls are a critical but often overlooked layer of defense. Bypassing network-layer security is unnecessary if physical access to server hardware is achievable — a reminder that security must be comprehensive across all layers.

Phase 4 — Exposed Personal / Contact Details

This phase documents any personal or contact information found exposed through WHOIS records, DNS TXT records, public web pages, or OSINT aggregation tools. Exposed personal data can enable social engineering, phishing, and targeted attacks.

4.1 WHOIS Contact Information

WHOIS records for many domains contain registrant contact details including names, email addresses, phone numbers, and postal addresses. If privacy protection (WHOIS guard) is not enabled, this data is freely accessible to anyone performing a WHOIS lookup.

For the target domain, WHOIS data was queried. Any personal identifiers found in these records are redacted in this public version of the report and replaced with [REDACTED] to prevent unintended exposure via this document.

4.2 Exposure Risk Assessment

Data Type	Source	Risk
IP Address	DNS resolution	Medium — CDN masked
Hosting Provider	WHOIS / ASN lookup	Low — public info
Server Location	Reverse DNS	Low — general region
Admin Contact (if any)	WHOIS record	**High — enables phishing**
Email Addresses	WHOIS / web crawl	**High — targeted spam/phishing**
ASN / Network Range	WHOIS IP lookup	Low — general infra info

Summary of Findings

The following table consolidates all significant findings from this reconnaissance exercise:

Target Domain	[Target Domain]
IP Address	18.66.57.84
Network	18.66.56.0/23
Hosting	AWS CloudFront CDN — Mumbai (bom78)
ASN	AS16509 (Amazon.com, Inc.)
Hostname	server-18-66-57-84.bom78.r.cloudfront.net
Origin IP Exposed?	No — masked by CloudFront CDN
SSL/TLS	Not verified in this phase
Physical Security	CCTV, access cards, restricted server room
WHOIS Privacy	Not confirmed — potential exposure risk

Recommendations

- **Enable WHOIS Privacy Protection:** Prevent registrant contact data exposure via public WHOIS lookup.
- **Verify SSL/TLS Configuration:** Ensure TLS 1.2+ only; disable SSLv2/v3 and weak cipher suites.
- **Review DNS Records:** Audit TXT, MX, and SPF records for misconfigurations or sensitive leaks.
- **Enforce DMARC/DKIM/SPF:** Prevent email spoofing from the domain — important for phishing defense.
- **Conduct Regular Recon Audits:** Periodically run OSINT checks to identify newly exposed data.
- **Strengthen Password Policy:** Adopt NIST SP 800-63B — prefer 12+ character passphrases.

Conclusion

This OSINT and reconnaissance exercise demonstrated that a significant amount of infrastructure information is publicly accessible through passive, non-intrusive methods. Without touching any production system or performing a single unauthorized action, the following was established: the target's IP range, hosting provider, CDN architecture, server geolocation, ASN, and inferred security policies.

For defenders, this highlights the importance of regularly auditing your own public footprint — an attacker's first move is always reconnaissance, and the more they can learn passively, the better they can craft targeted attacks. For security professionals, OSINT is a foundational skill that underpins every subsequent phase of a penetration test.

End of Report